



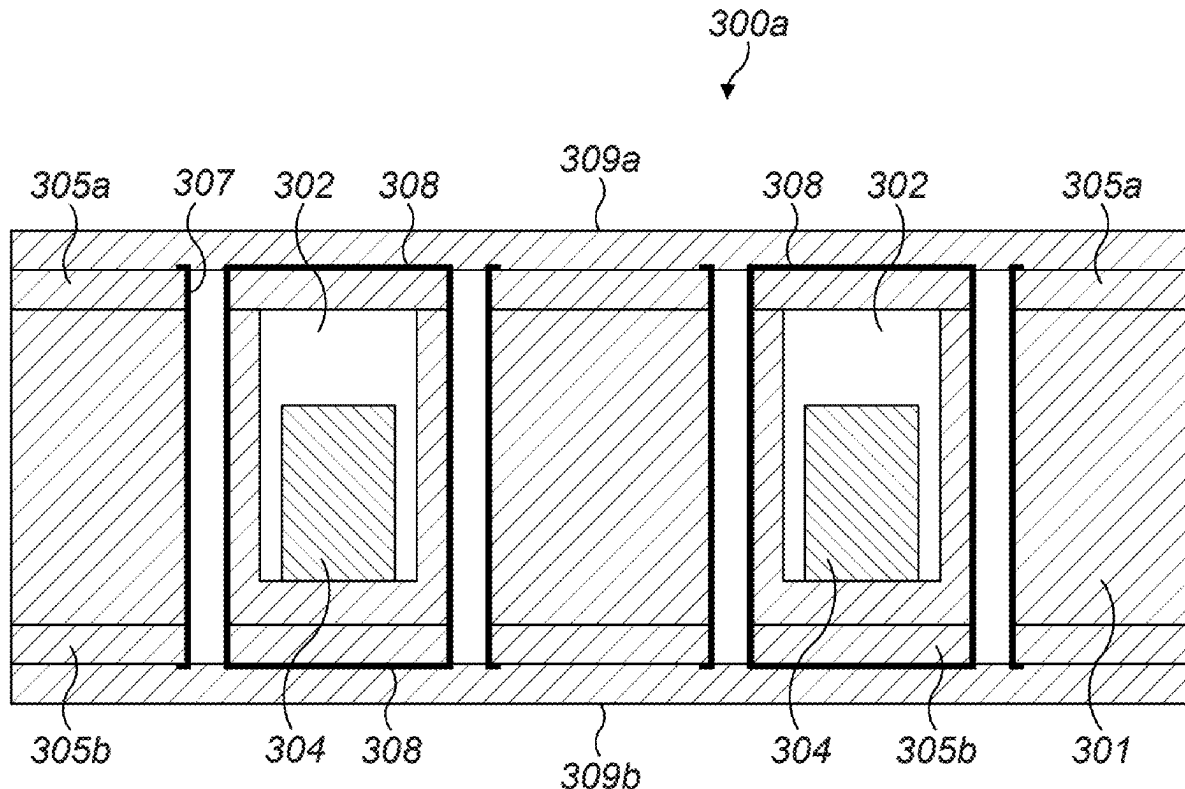
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(19) **United States**(12) **Patent Application Publication**
PARISH et al.(10) **Pub. No.: US 2025/0285798 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **METHOD OF MANUFACTURING AN
EMBEDDED MAGNETIC COMPONENT
DEVICE**(71) Applicant: **Murata Manufacturing Co., Ltd.**,
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continuation of application No. 16/852,588, filed on
Apr. 20, 2020, now Pat. No. 11,894,176, which is a
division of application No. 15/717,998, filed on Sep.
28, 2017, now Pat. No. 10,665,379, which is a division
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2015, now Pat. No. 10,176,917.**Foreign Application Priority Data**

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(2013.01); **H01F 2027/2809** (2013.01); **H01F**
41/125 (2013.01)(57) **ABSTRACT**

An embedded magnetic component device includes a magnetic core located in a cavity extending into an insulating substrate. The cavity and magnetic core are covered with a cover layer. Through holes extend through the cover layer and the insulating substrate, and are plated to define conductive vias. Metallic traces are provided at exterior surfaces of the cover layer and the insulating substrate to define upper and lower winding layers. The metallic traces and conductive vias define the respective primary and secondary side windings for an embedded transformer. At least a first isolation barrier is provided on the cover layer, and at least a third insulating layer is provided on the substrate. The second and third insulating layers provide additional insulation for the device, and define and function as a circuit board for surface mounted power electronics.



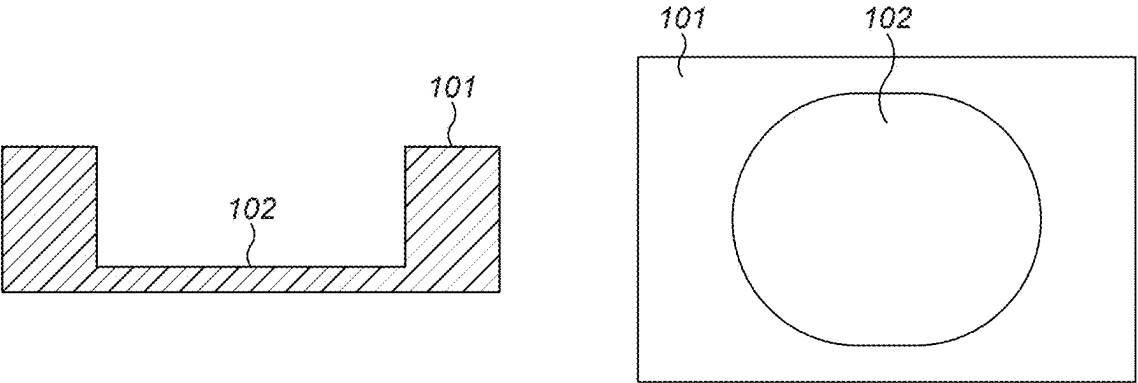


FIG. 1A
(Prior Art)

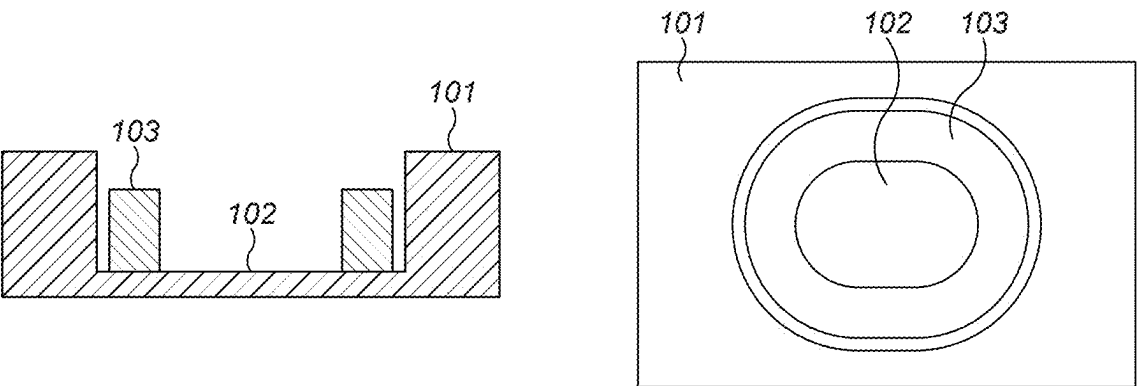


FIG. 1B
(Prior Art)

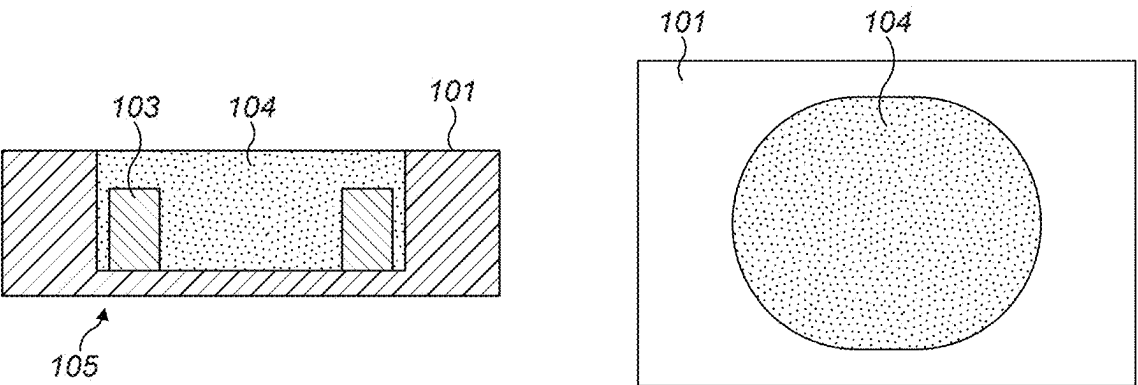


FIG. 1C
(Prior Art)

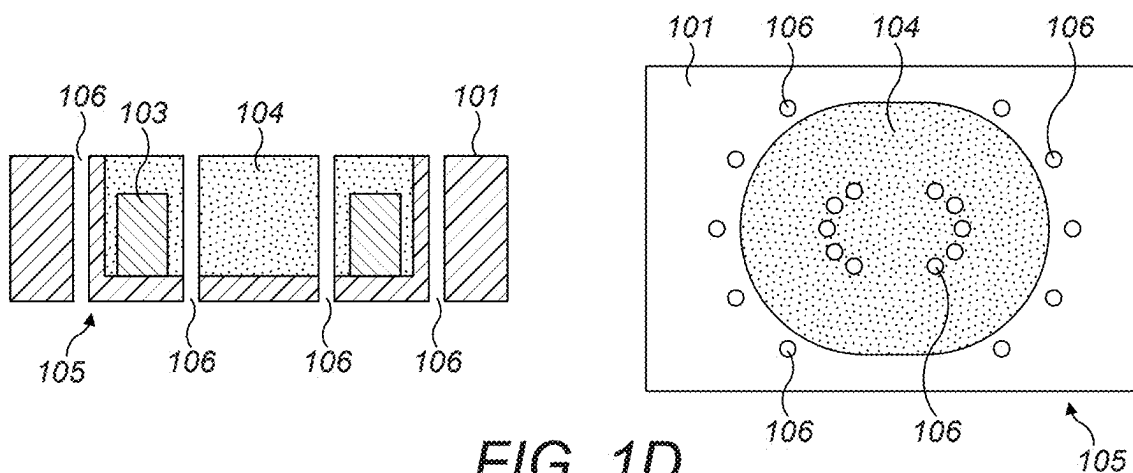


FIG. 1D
(Prior Art)

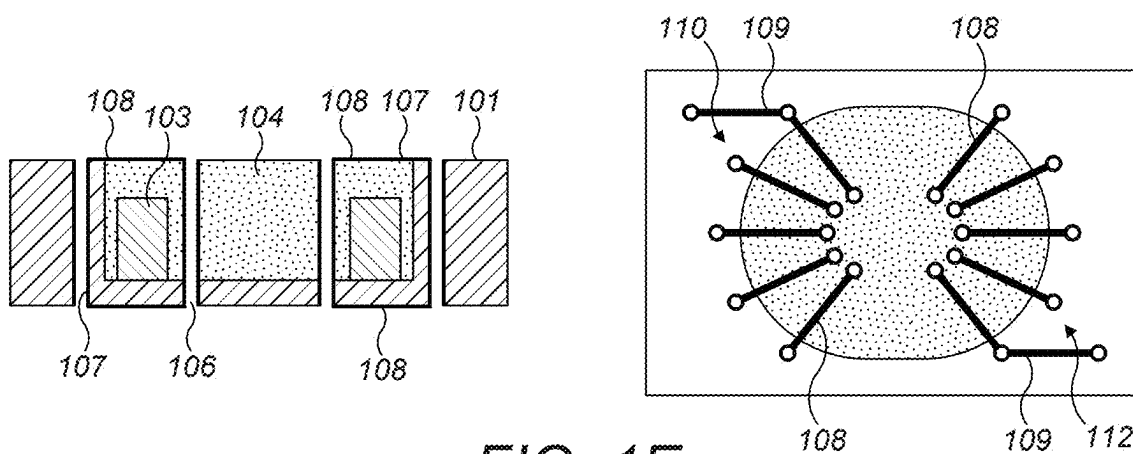


FIG. 1E
(Prior Art)

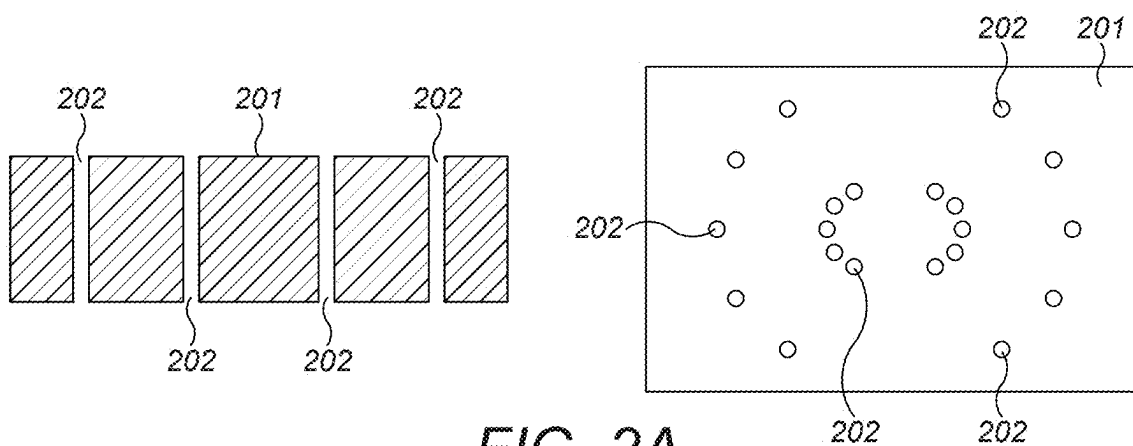


FIG. 2A
(Prior Art)

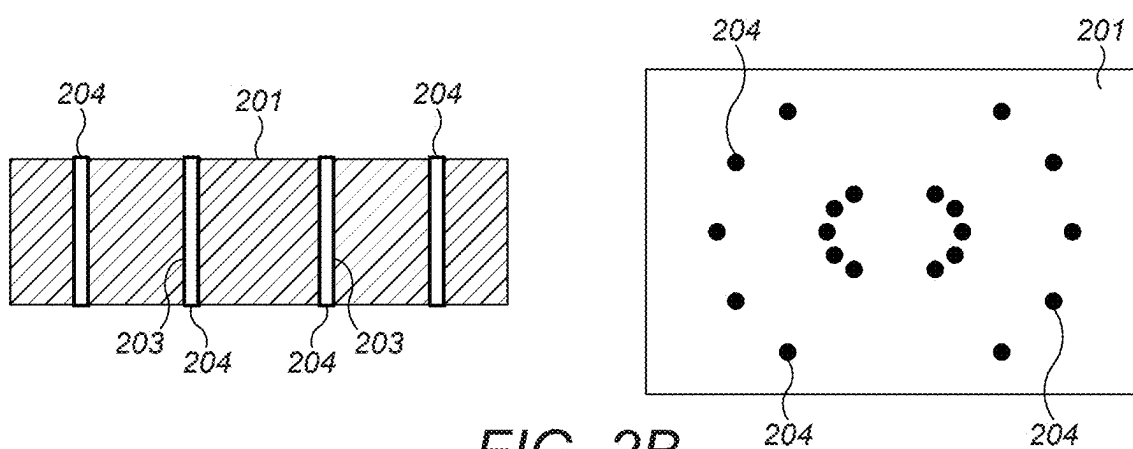


FIG. 2B
(Prior Art)

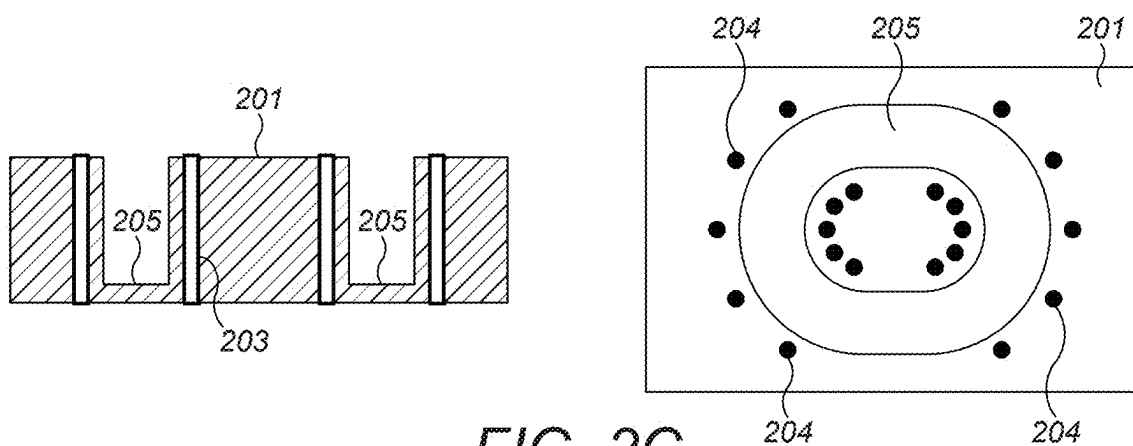


FIG. 2C
(Prior Art)

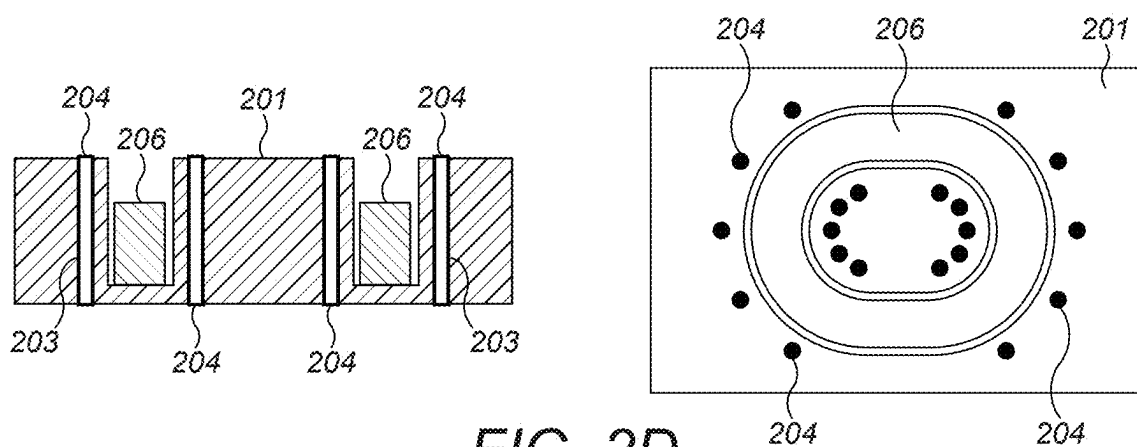


FIG. 2D
(Prior Art)

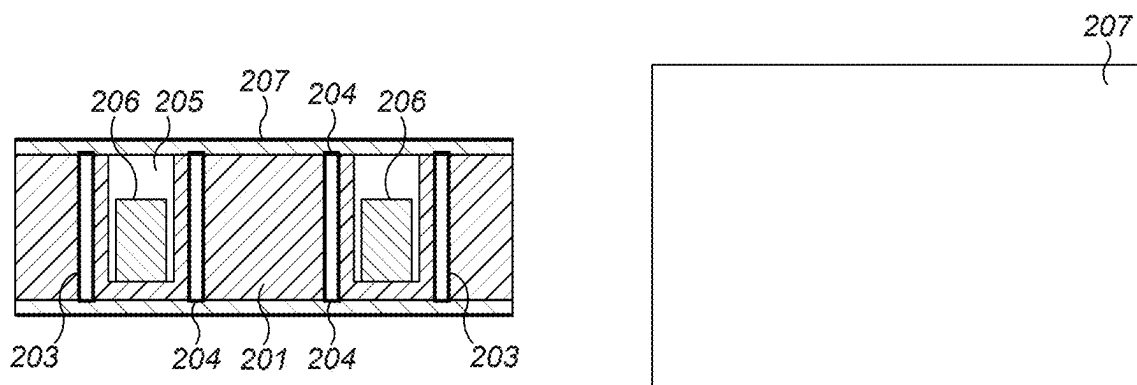


FIG. 2E
(Prior Art)

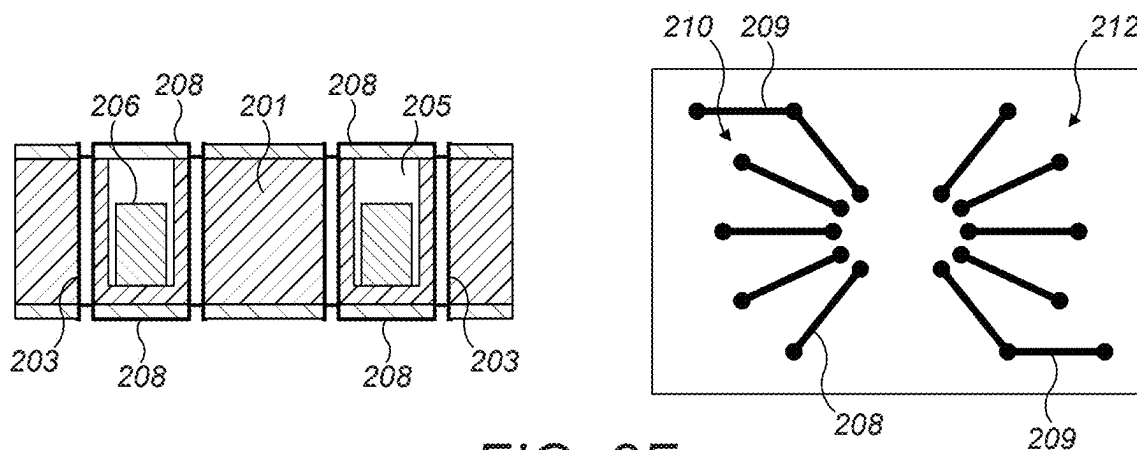


FIG. 2F
(Prior Art)

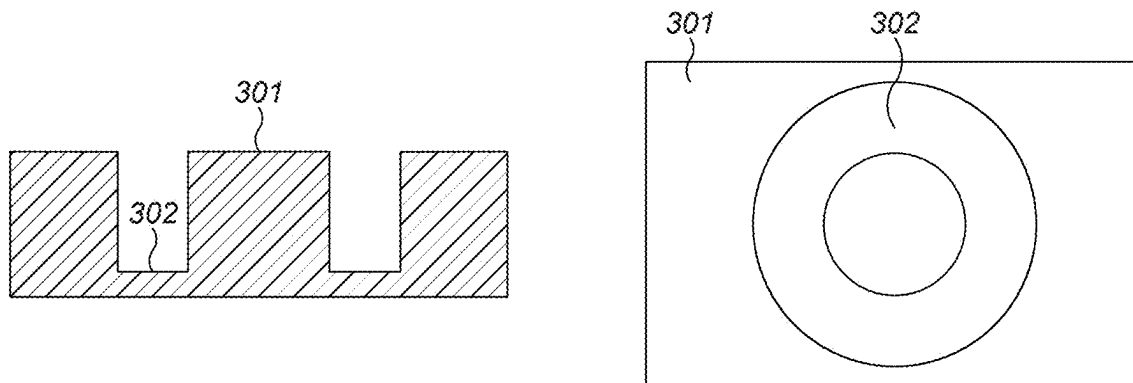


FIG. 3A

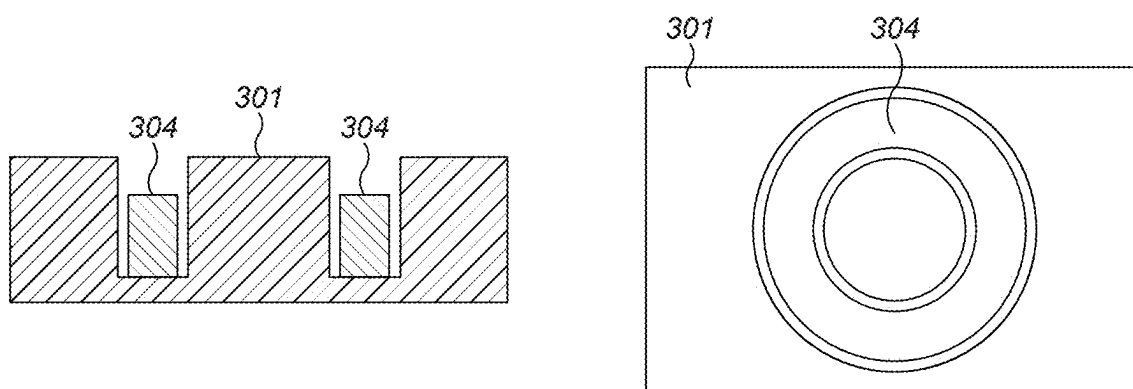


FIG. 3B

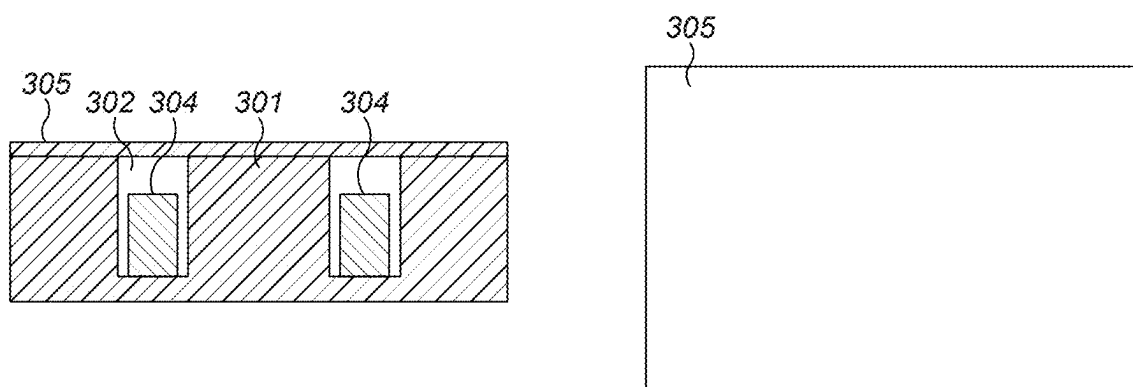


FIG. 3C

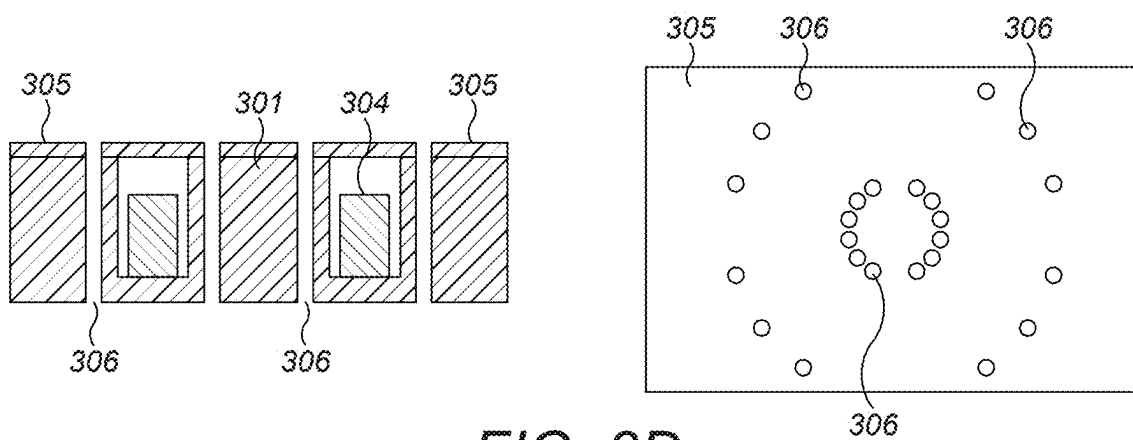


FIG. 3D

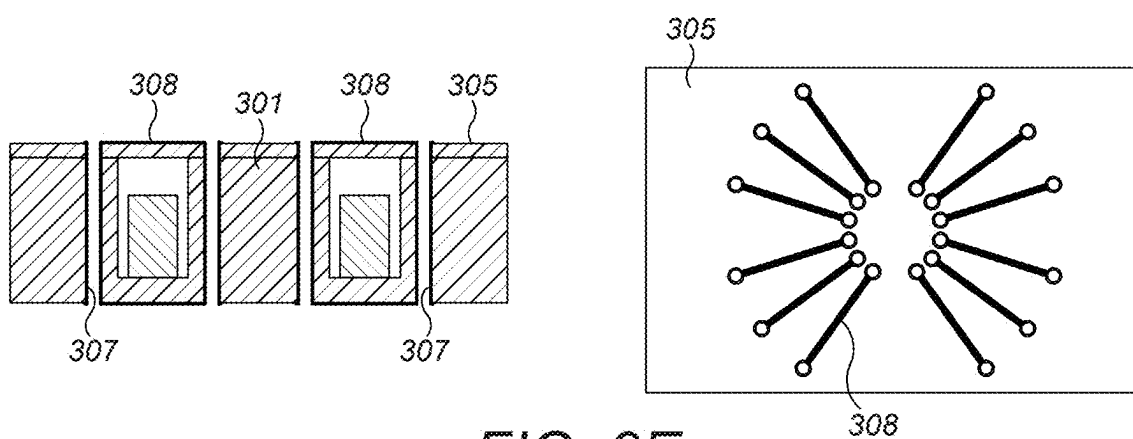


FIG. 3E

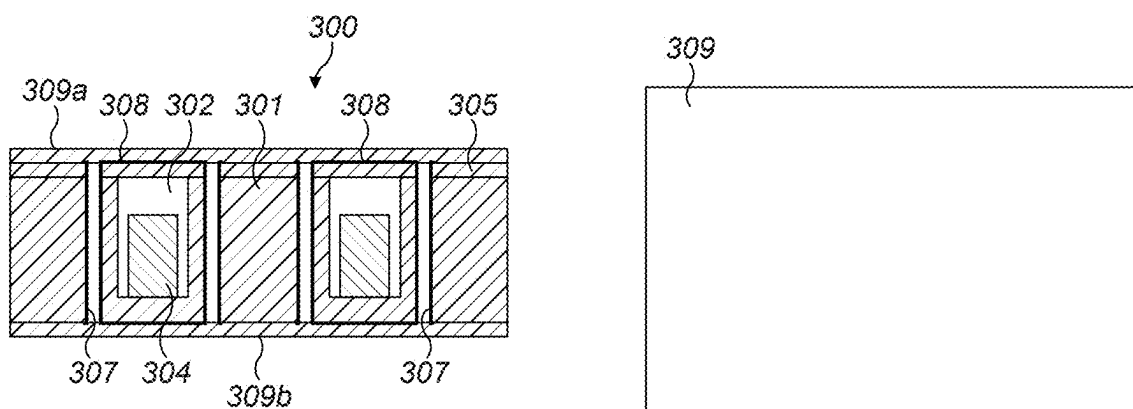


FIG. 3F

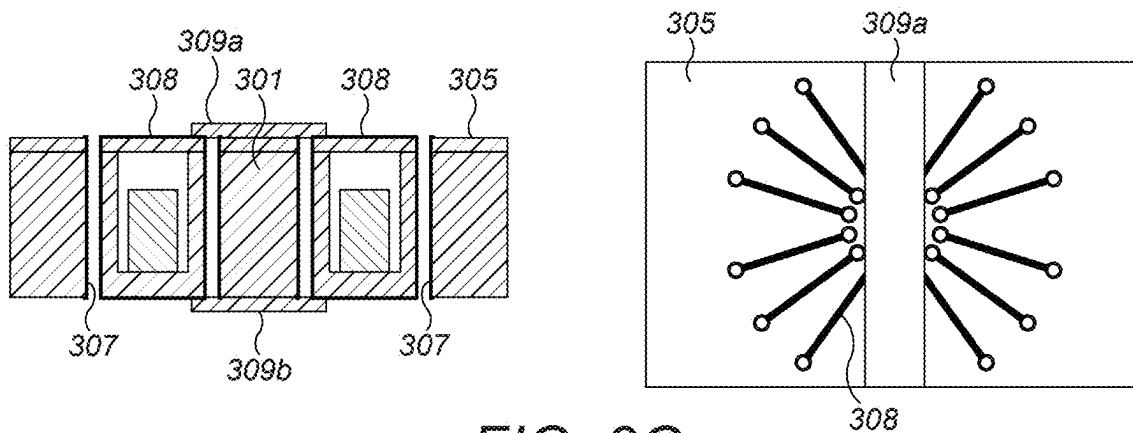


FIG. 3G

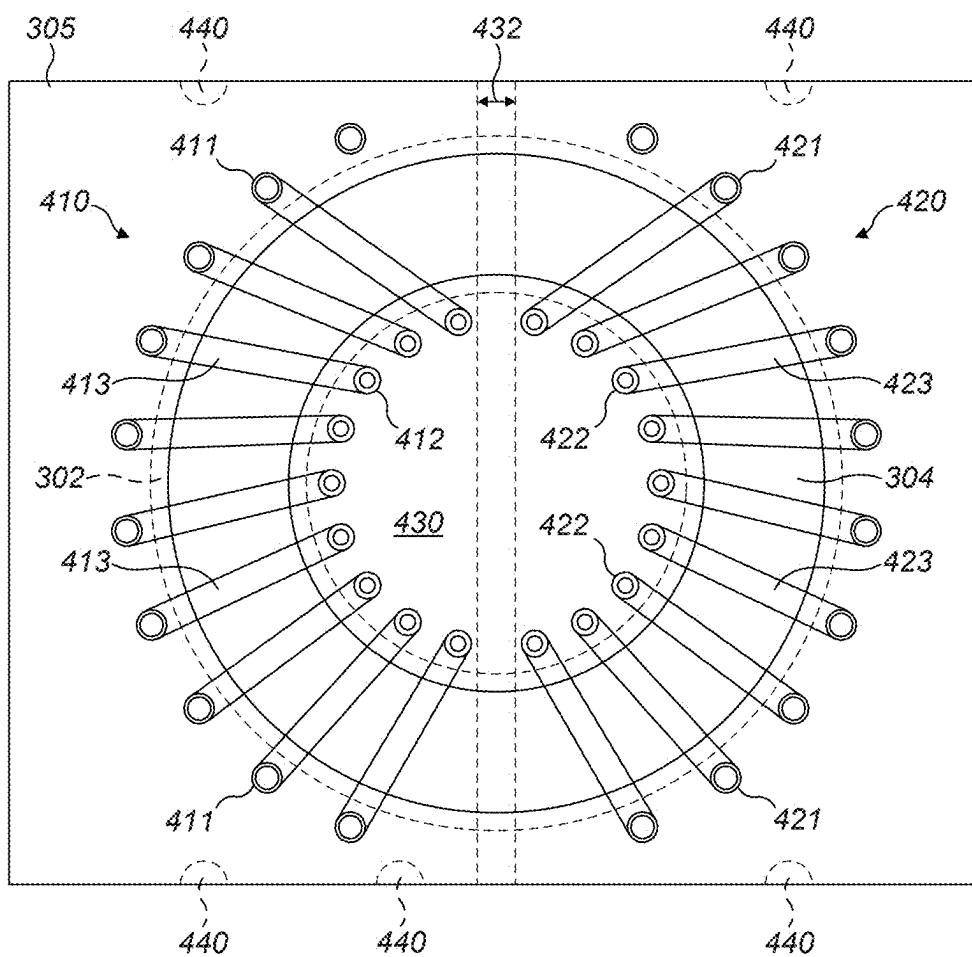


FIG. 4A

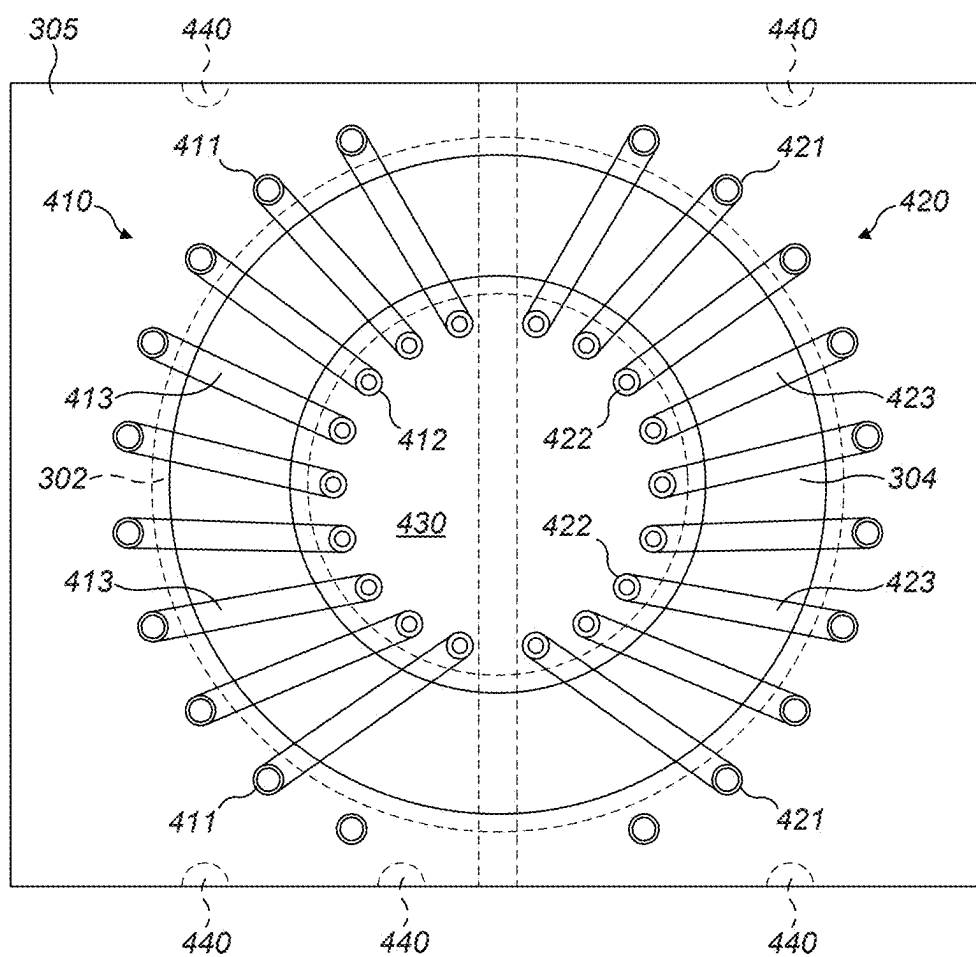


FIG. 4B

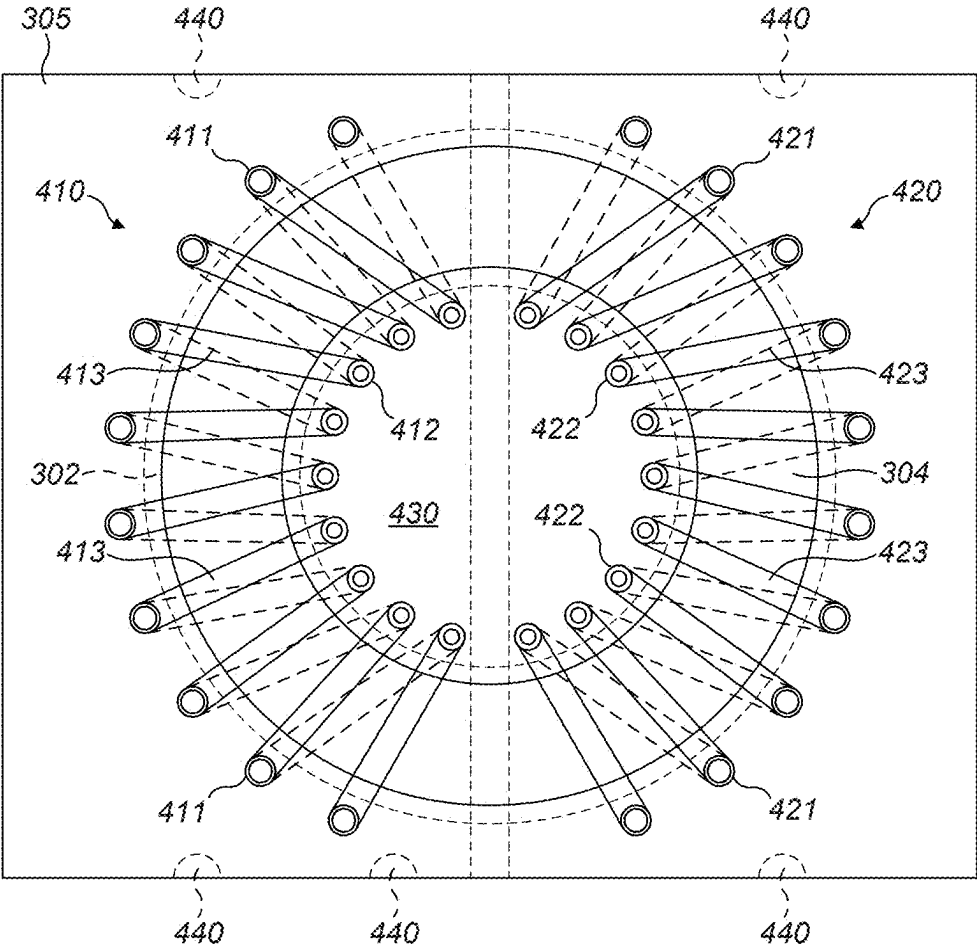


FIG. 4C

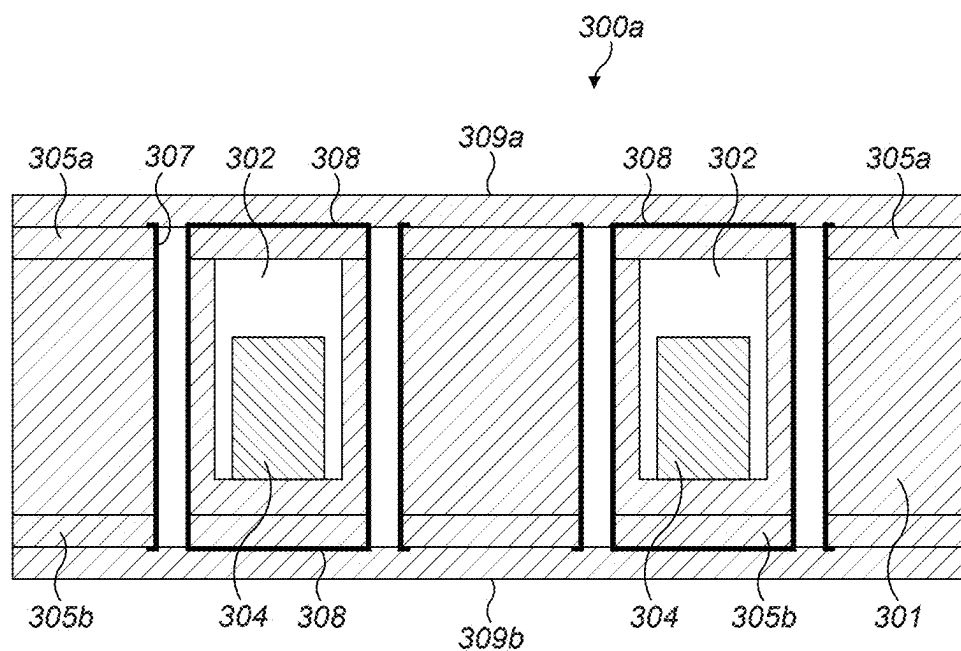


FIG. 5

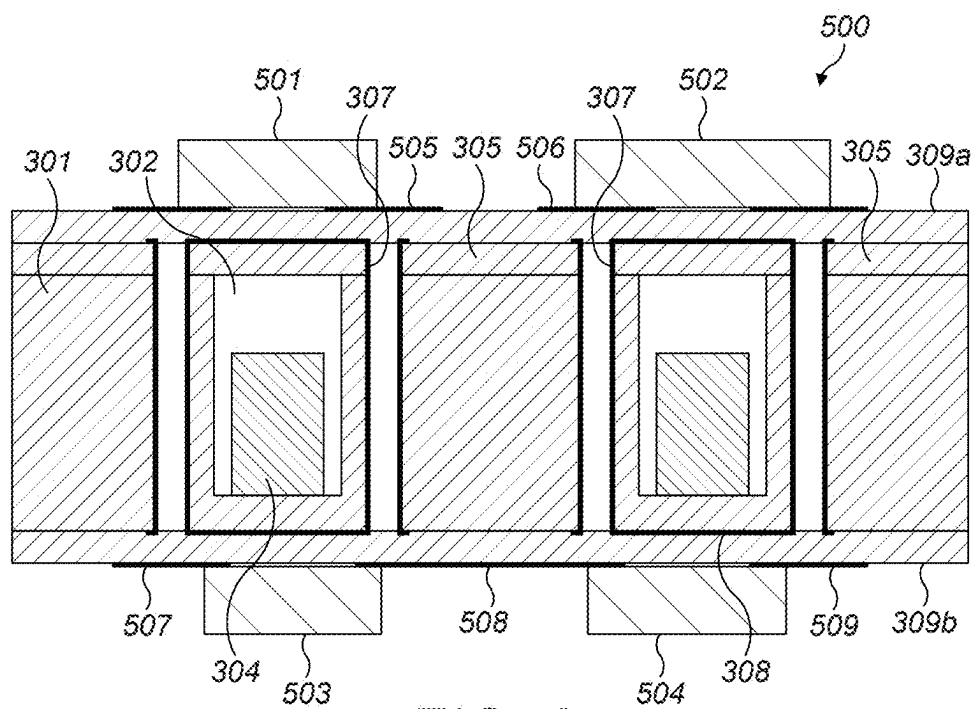


FIG. 6

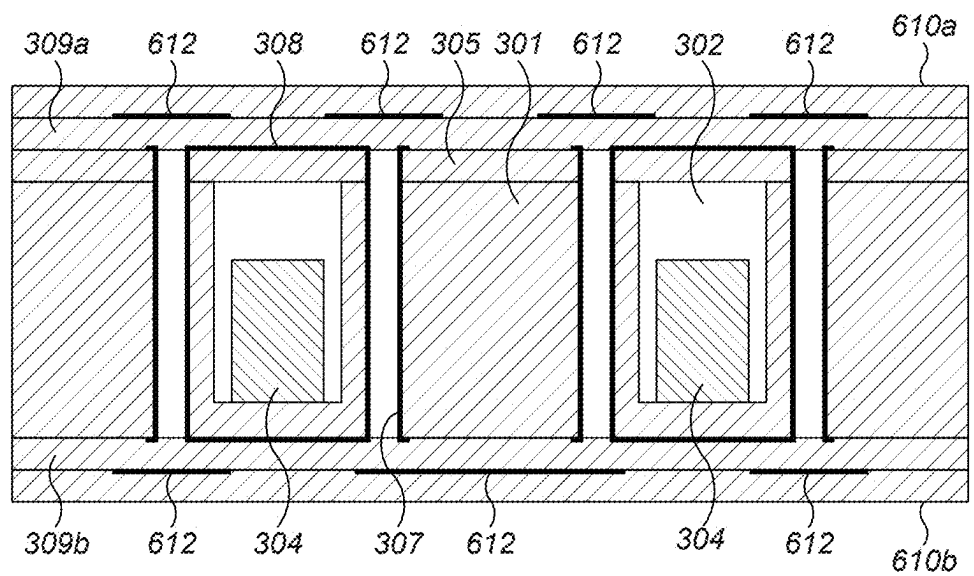


FIG. 7

METHOD OF MANUFACTURING AN EMBEDDED MAGNETIC COMPONENT DEVICE

BACKGROUND OF THE INVENTION

1. Field of the Invention

[0001] The present invention relates to embedded magnetic component devices, and in particular to embedded magnetic component devices with improved isolation performance.

2. Description of the Related Art

[0002] Power supply devices, such as transformers and converters, involve magnetic components such as transformer windings and often magnetic cores. The magnetic components typically contribute the most to the weight and size of the device, making miniaturization and cost reduction difficult.

[0003] In addressing this problem, it is known to provide low-profile transformers and inductors in which the magnetic components are embedded in a cavity in a resin substrate, and the necessary input and output electrical connections for the transformer or inductor are formed on the substrate surface. A printed circuit board (PCB) for a power supply device can then be formed by adding layers of solder resist and copper plating to the top and/or bottom surfaces of the substrate. The necessary electronic components for the device may then be surface mounted on the PCB. This allows a significantly more compact and thinner device to be built.

[0004] In US2011/0108317, for example, a packaged structure having a magnetic component that can be integrated into a printed circuit board, and a method for producing the packaged structure, are described. In a first method, illustrated in FIGS. 1A to 1E, an insulating substrate 101, made of epoxy based glass fiber, has a cavity 102 (FIG. 1A). An elongate toroidal magnetic core 103 is inserted into the cavity 102 (FIG. 1B), and the cavity is filled with an epoxy gel 104 (FIG. 1C) so that the magnetic component 103 is fully covered. The epoxy gel 104 is then cured, forming a solid substrate 105 having an embedded magnetic core 103.

[0005] Through-holes 106 for forming primary and secondary side transformer windings are then drilled in the solid substrate 105 on the inside and outside circumferences of the toroidal magnetic component 103 (FIG. 1D). The through-holes 106 are then plated with copper, to form vias 107, and metallic traces 108 are formed on the top and bottom surfaces of the solid substrate 105 to connect respective vias 107 together into a winding configuration (FIG. 1E) and to form input and output terminals 109. In this way, a coil conductor is created around the magnetic component. The coil conductor shown in FIG. 1E is for an embedded transformer and has left and right coils forming primary and secondary side windings. Embedded inductors can be formed in the same way, but may vary in terms of the input and output connections, the spacing of the vias, and the type of magnetic core used.

[0006] A solder resist layer can then be added to the top and bottom surfaces of the substrate covering the metallic surface terminal lines, allowing further electronic components to be mounted on the solder resist layer. In the case of

power supply converter devices, for example, one or more transistor switching devices and associated control electronics, such as Integrated Circuit (ICs) and passive components, may be mounted on the surface resist layer.

[0007] Devices manufactured in this way have a number of associated problems. In particular, air bubbles may form in the epoxy gel 104 as it is solidifying. During reflow soldering of the electronic components on the surface of the substrate, these air bubbles can expand and cause failure in the device.

[0008] US2011/0108317 also describes a second technique in which epoxy gel is not used to fill the cavity. This second technique will be described with respect to FIGS. 2A to 2E.

[0009] As illustrated in FIG. 2A, through-holes 202 are first drilled into a solid resin substrate 201 at locations corresponding to the interior and exterior circumference of an elongate toroidal magnetic core. The through-holes 202 are then plated to form the vertical conductive vias 203 of the transformer windings, and metallic caps 204 are formed on the top and the bottom of the conductive vias 203 as shown in FIG. 2B. A toroidal cavity 205 for the magnetic core is then routed in the solid resin substrate 201 between the conductive vias 203 (FIG. 2C), and a ring-type magnetic core 206 is placed in the cavity 205 (FIG. 2D). The cavity 205 is slightly larger than the magnetic core 206, and an air gap may therefore exist around the magnetic core 206.

[0010] Once the magnetic core 206 has been inserted into the cavity 205 an upper epoxy dielectric layer 207 (such as an adhesive bondply layer) is added to the top of the structure, to cover the cavity 205 and the magnetic core 206. A corresponding layer 207 is also added to the bottom of the structure (FIG. 2E) on the base of the substrate 201. Further through-holes are drilled through the upper and lower epoxy layers 207 to the caps 204 of the conductive vias 203, and plated, and metallic traces 208 are subsequently formed on the top and bottom surfaces of the device as before (FIG. 2F), to form input and output terminals 209.

[0011] As noted above, where the embedded magnetic components of FIGS. 1 and 2 are transformers, a first set of windings 110, 210 provided on one side of the toroidal magnetic core form the primary transformer coil, and a second set of windings 112, 212 on the opposite side of the magnetic core form the secondary windings. Transformers of this kind can be used in power supply devices, such as isolated DC-DC converters, in which isolation between the primary and secondary side windings is required. In the example devices illustrated in FIGS. 1 and 2, the isolation is a measure of the minimum spacing between the primary and secondary windings.

[0012] In the case of FIGS. 1 and 2 above, the spacing between the primary and secondary side windings must be large to achieve a high isolation value, because the isolation is only limited by the dielectric strength of the air, in this case in the cavity or at the top and bottom surfaces of the device. The isolation value may also be adversely affected by contamination of the cavity or the surface with dirt.

[0013] For many products, safety agency approval is required to certify the isolation characteristics. If the required isolation distance though air is large, there will be a negative impact on product size. For mains reinforced voltages (250 Vrms), for example, a spacing of approximately 5 mm is required across a PCB from the primary

windings to the secondary windings in order to meet the insulation requirements of EN/UL60950.

[0014] It would be desirable to provide an embedded magnetic component device with improved isolation characteristics, and to provide a method for manufacturing such a device.

SUMMARY OF THE INVENTION

[0015] A preferred embodiment of the present invention provides an embedded magnetic component device, including a magnetic core made of ferrite and including a first section and a second section; an insulating substrate made of a resin material and including a first side and second side opposite to the first side, the insulating substrate including a cavity housing the magnetic core with an air gap between the cavity and the magnetic core; a primary electrical winding passing through the insulating substrate and disposed around the first section of the magnetic core, the primary electrical winding is located on the first and the second sides of the insulating substrate; a secondary electrical winding passing through at least the insulating substrate, disposed around the second section of the magnetic core, and spaced away from the primary electrical winding so as to be isolated from the primary electrical winding, the secondary electrical winding is located on the first and the second sides of the insulating substrate; a first isolation barrier made of a resin material which is not a solder resist, located on the first side of the insulating substrate, covering at least the portion of the first side between the primary electrical winding and the second electrical winding where the primary electrical winding and the second electrical winding are closest, and including a solid bonded joint with the first side of the insulating substrate; a second isolation barrier made of a resin material which is not a solder resist, located on the second side of the insulating substrate, covering at least a portion of the second side between the primary electrical winding and the second electrical winding where the primary electrical winding and the second electrical winding are closest, and including a solid bonded joint with the second side of the insulating substrate. The primary electrical winding includes first upper conductive traces disposed on the first side of the insulating substrate and covered by the first isolation barrier; first lower conductive traces disposed on the second side of the insulating substrate and covered by the second isolation barrier; first inner conductive connectors disposed in the insulating substrate near the inner periphery of the magnetic core and providing an electrical connection between the first upper conductive traces and the first lower conductive traces; and first outer conductive connectors disposed in the insulating substrate near the outer periphery of the magnetic core and providing an electrical connection between the first upper conductive traces and the first lower conductive traces, and the secondary electrical winding includes second upper conductive traces disposed on the first side of the insulating substrate and covered by the first isolation barrier; second lower conductive traces disposed on the second side of the insulating substrate and covered by the second isolation barrier; second inner conductive connectors disposed in the insulating substrate near the inner periphery of the magnetic core and providing an electrical connection between the second upper conductive traces and the second lower conductive traces, and second outer conductive connectors disposed in the insulating substrate near the outer periphery of the magnetic core and providing an electrical

connection between the second upper conductive traces and the second lower conductive traces.

[0016] The first isolation barrier may include a single layer and may cover the first side of the insulating substrate entirely.

[0017] Alternatively, the first isolation barrier may include a plurality of layers and may cover the first side of insulating substrate entirely.

[0018] The second isolation barrier may include a single layer and may cover the second side of the insulating substrate entirely.

[0019] The second isolation barrier may include a plurality of layers and may cover the second side of insulating substrate entirely.

[0020] The device may include a first solder resist layer covering the first isolation barrier, and a second solder resist layer covering the second isolation barrier.

[0021] The insulating substrate may include a base insulating substrate and a cover layer located on the base substrate, the cover layer covering the cavity in which the magnetic core is housed and providing the first surface of the insulating substrate.

[0022] The device may further include first land patterns located on the surface of the first isolation barrier that is opposite to the side covering the primary electrical winding and the second electrical winding, and electronic components mounted on the first land patterns.

[0023] The device may further include second land patterns located on the surface of the second isolation barrier that is opposite to the side covering the primary electrical winding and the second electrical winding, and electronic components mounted on the second land patterns.

[0024] The insulating substrate, the first isolation barrier, and the second isolation barrier may be made of the same material.

[0025] The first isolation barrier and the second isolation barrier may include a thermoplastic, a ceramic material, or an epoxy material.

[0026] Another preferred embodiment of the present invention provides a method of manufacturing an embedded magnetic component device, including a) preparing an insulating substrate formed of a resin material and including a cavity therein; b) installing a magnetic core made of ferrite in the cavity with an air gap between the cavity and the magnetic core, the magnetic core including a first section and a second section; c) forming isolated primary and secondary electrical windings that pass through the insulating substrate and that are disposed around the first and second sections of the magnetic core, the secondary winding being spaced away from the primary electrical winding so as to be isolated from the primary electrical winding, the primary electrical winding is located on the first and the second sides of the insulating substrate, and the secondary electrical winding is located on the first and the second sides of the insulating substrate; d) forming a first isolation barrier of a resin material which is not a solder resist on the first side of the insulating substrate, covering at least the portion of the first side of the insulating substrate between the primary electrical winding and the secondary electrical winding where the primary electrical winding and the secondary electrical winding are closest, to form a solid bonded joint with the first side of the insulating substrate; and e) forming a second isolation barrier of a resin material which is not a solder resist on the second side of the insulating substrate,

covering at least the portion of the second side of the insulating substrate between the primary electrical winding and the secondary electrical winding where the primary electrical winding and the secondary electrical winding are closest, to form a solid bonded joint with the second side of the insulating substrate. Step d) includes forming upper conductive traces on the first side of the insulating substrate; forming lower conductive traces on the second side of the insulating substrate; forming through holes through the insulating substrate; plating the through holes to form conductive vias to connect the upper conductive traces and the lower conductive traces and to form a coil conductor around the magnetic core. The first isolation barrier covers the upper conductive traces, and the second isolation barrier covers the lower conductive traces.

[0027] Step d) may alternatively include laminating a plurality of insulating layers to form the first and/or second isolation barrier, wherein the first and second isolation barriers entirely cover the first and second sides of the insulating substrate.

[0028] The insulating substrate may include a base insulating substrate and a cover layer formed on the base insulating substrate, and the cover layer may cover the cavity in which the magnetic core is housed and may provide the first side of the insulating substrate.

[0029] Step c) may include laminating an additional insulating layer on the base insulating substrate opposite to the cover layer.

[0030] The method may further include forming first land patterns on the surface of the first isolation barrier that is opposite to the side covering the primary electrical winding and the second electrical winding; and mounting electronic components on the first land patterns.

[0031] The method may further include forming second land patterns on the surface of the second isolation barrier that is opposite to the side covering the primary electrical winding and the second electrical winding; and mounting electronic components on the second land patterns.

[0032] The method may further include forming a first solder resist layer covering the first isolation barrier; and forming a second solder resist layer covering the second isolation barrier.

[0033] The insulating substrate, the cover layer, and the first and second isolation barriers may be made of the same material.

[0034] The insulating substrate may include a thermoplastic, a ceramic material, or an epoxy material.

[0035] The above and other features, elements, characteristics, steps, and advantages of the present invention will become more apparent from the following detailed description of preferred embodiments of the present invention with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0036] FIGS. 1A to 1E illustrate a first known technique for manufacturing a substrate including an embedded magnetic component.

[0037] FIGS. 2A to 2F illustrate a second known technique for manufacturing a substrate including an embedded magnetic component.

[0038] FIGS. 3A to 3F show a technique for manufacturing the device according to a first preferred embodiment of the present invention.

[0039] FIG. 3G shows a variation on the device shown in FIG. 3F.

[0040] FIG. 4A illustrates a top down view of the cavity, the magnetic core, and the conductive vias; FIG. 4B illustrates the reverse side of the device and cavity; and FIG. 4C is a schematic illustration of the conductive vias showing the trace pattern connecting adjacent vias together to define the windings.

[0041] FIG. 5 illustrates a second preferred embodiment of the present invention.

[0042] FIG. 6 illustrate a third preferred embodiment of the present invention, incorporating the embedded magnetic component device of FIG. 3A-3F or 5 into a larger device.

[0043] FIG. 7 illustrates a fourth preferred embodiment of the present invention including additional layers of insulating material.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

First Preferred Embodiment

[0044] A first preferred embodiment of the present invention of an embedded magnetic component device will now be described with reference to FIGS. 3A to 3F. A completed embedded magnetic component device according to the first preferred embodiment is illustrated in FIG. 3F.

[0045] In a first step, illustrated in FIG. 3A, a circular annulus or cavity 302 for housing a magnetic core is routed in an insulating substrate 301. In this preferred embodiment, the insulating substrate 301 is formed of a resin material, such as FR4. FR4 is a composite 'pre-preg' material composed of woven fiberglass cloth impregnated with an epoxy resin binder. The resin is pre-dried, but not hardened, so that when it is heated, it flows and acts as an adhesive for the fiberglass material. FR4 has been found to have favourable thermal and insulation properties.

[0046] As shown in FIG. 3B, a circular magnetic core 304 is then installed in the cavity 302. The cavity 302 may be slightly larger than the magnetic core 304, so that an air gap may exist around the magnetic core 304. The magnetic core 304 may be installed in the cavity manually or by a surface mounting device such as a pick and place machine.

[0047] In the next step, illustrated in FIG. 3C, a first insulating layer 305 or cover layer is secured or laminated on the insulating substrate 301 to cover the cavity 302 and the magnetic core 304. Preferably, the cover layer 305 is formed of the same material as the insulating substrate 301 as this aids bonding between the top surface of the insulating substrate 301 and the lower surface of the cover layer 305. The cover layer 305 may therefore also be formed of a material such as FR4, laminated onto the insulating substrate 301. Lamination may be via adhesive or via heat activated bonding between layers of pre-preg material. In other preferred embodiments, other materials may be used for the cover layer 305.

[0048] In the next step illustrated in FIG. 3D, through-holes 306 are formed through the insulating substrate 301 and the cover layer 305. The through holes 306 are formed at suitable locations to form the primary and secondary coil conductor windings of an embedded transformer. In this preferred embodiment, as the transformer has the magnetic core 304 that is round or circular in shape, the through-holes 306 are therefore suitably formed along sections of two arcs corresponding to inner and outer circular circumferences. As

is known in the art, the through-holes **306** may be formed by drilling, or other suitable technique. A schematic illustration of an example pattern of conductive vias is shown in FIGS. 4A-4C and described below.

[0049] As shown in FIG. 3E, the through-holes **306** are then plated to form conductive via holes **307** that extend from the top surface of the cover layer to the bottom surface of the substrate **301**. Conductive or metallic traces **308** are added to the top surface of the cover layer **305** to form an upper winding layer connecting the respective conductive via holes **307**, and in part forming the windings of the transformer. The upper winding layer is illustrated by way of example in the right hand side of FIG. 3E. The metallic traces **308** and the plating for the conductive via holes **307** are usually formed from copper, and may be formed in any suitable way, such as by adding a copper conductor layer to the outer surfaces of the layer **305** which is then etched to form the necessary patterns, deposition of the copper onto the surface, and so on.

[0050] Metallic traces **308** are also formed on the bottom surface of the insulating substrate **301** to form a lower winding layer also connecting the respective conductive via holes **307** to partly form the windings of the transformer. The upper and lower winding layers **308** and the via holes **307** together form the primary and secondary windings of the transformer.

[0051] Lastly, as shown in FIG. 3F, second and third insulating layers **309** are formed on the top and bottom surfaces of the structure shown in FIG. 3E to form first and second isolation barriers. The layers may be secured in place by lamination or other suitable technique. The bottom surface of the second insulating layer or first isolation barrier **309a** adheres to the top surface of the cover layer **305** and covers the terminal lines of the upper winding layer **308**. The top surface of the third insulating layer or second isolation barrier **309b** on the other hand adheres to the bottom surface of the substrate **301** and so covers the terminal lines of the lower winding layer **308**. Advantageously, the second and third insulating layers, i.e., first isolation barrier **309a** and second isolation barrier **309b**, may also be formed of FR4, and so laminated onto the insulating substrate **301** and cover layer **305** using the same process as for the cover layer **305**.

[0052] Through holes and via conductors are formed through the second and third insulating layers, i.e., first isolation barrier **309a** and second isolation barrier **309b**, in order to connect to the input and output terminals of the primary and second transformer windings (not shown). Where the vias through the second and third insulating layers, i.e., first isolation barrier **309a** and second isolation barrier **309b**, are located apart from the vias through the substrate **301** and the cover layer **305**, a metallic trace will be needed on the upper winding layer connecting the input and output vias to the first and last via in each of the primary and secondary windings. Where the input and output vias are formed in overlapping positions, then conductive or metallic caps could be added to the first and last via in each of the primary and secondary windings.

[0053] FIG. 3F illustrates a finished embedded magnetic component device **300** according to a first preferred embodiment of the present invention. The first and second isolation barriers **309a** and **309b** form a solid bonded joint with the adjacent layers, either cover layer **305** or substrate **301**, on which the upper or lower winding layers **308** of the transformer are formed. The first and second isolation barriers

309a and **309b** therefore provide a solid insulated boundary along the surfaces of the embedded magnetic component device, greatly reducing the chance of arcing or breakdown, and allowing the isolation spacing between the primary and secondary side windings to be greatly reduced.

[0054] To meet the insulation requirements of EN/UL60950 only 0.4 mm is required through a solid insulator for mains referenced voltages (250 Vrms).

[0055] The first and second isolation barriers **309a** and **309b** are formed on the substrate **301** and cover layer **305** without any air gap between the layers. If there is an air gap in the device, such as above or below the winding layers, then there would be a risk of arcing and failure of the device. The first and second isolation barriers **309a** and **309b**, the cover layer **305** and the substrate **301**, therefore form a solid block of insulating material.

[0056] In the above-described figures, the first and second isolation barriers **309a** and **309b** are illustrated as covering the whole of the cover layer **305** and the bottom surface of the substrate **301** of the embedded magnetic component device **300**. In alternative preferred embodiments, however, it may be sufficient if the first and second isolation barriers are applied to the cover layer **305** and the bottom of the substrate **301** so that they at least cover only the portion of the surface of the cover layer **305** and substrate **301** surface between the primary and secondary windings, where the primary and secondary windings are closest. In FIG. 3G for example, the first and second isolation barriers **309a** and **309b** may be provided as a long strip of insulating material placed on the surface parallel or substantially parallel to the shorter edge of the device and covering at least the isolation region **430** (see FIGS. 4A-4C below) between the primary and secondary side windings. In alternative preferred embodiments, as the primary and secondary side windings follow the arc of the magnetic core **304** around which they are wound, it may be sufficient to place the isolation barriers **309a** and **309b** only where the primary and secondary side windings are closest, which in this case is at the 12 o'clock and 6 o'clock positions. As noted above, however, a full layer of the first and second isolation barriers **309a** and **309b** covering the entire surface of the embedded component device can be advantageous as it provides locations for further mounting of components on the surface of the device.

[0057] The pattern of through holes **306**, conductive vias **307**, and metallic traces **308** forming the upper and lower winding layers of the transformer will now be described in more detail with reference to FIG. 4A. FIG. 4A is a top view of the embedded magnetic component device with the upper winding layer exposed. The primary windings **410** of the transformer are shown on the left hand side of the device, and the secondary windings **420** of the transformer are shown on the right hand side. One or more tertiary or auxiliary transformer windings may also be formed, using the conductive vias **307** and metallic traces **308** but are not illustrated in FIG. 4A. In FIG. 4A, input and output connections to the transformer windings are also omitted to avoid obscuring the detail.

[0058] The primary winding of the transformer **410** includes outer conductive vias **411** arranged around the outer periphery of the circular cavity **302** containing the magnetic core **304**. As illustrated in FIG. 4A, the outer conductive vias

411 closely follow the outer circumference or periphery of the cavity **302** and are arranged radially in a row, along a section of arc.

[0059] Inner conductive vias **412** are provided in the inner or central region of the substrate **301**. The inner conductive vias are arranged to closely follow the inner circumference or periphery of the cavity **302** and are arranged radially in a row, along a section of arc. Each outer conductive via **411** in the upper winding layer **308** is connected to a single inner conductive via **412** by a metallic trace **413**. The metallic traces **413** are formed on the surface of the cover layer **305** and so cannot overlap with one another. Although, the inner conductive vias **412** need not strictly be arranged in rows, it is helpful to do so, as an ordered arrangement of the inner conductive vias **412** assists in arranging the metallic traces **413** so that they connect the outer conductive vias **411** to the inner conductive vias **412**.

[0060] The secondary winding of the transformer **420** also includes outer conductive vias **421** and inner conductive vias **422** connected to each other by respective metallic traces **423** in the same way as for the primary winding.

[0061] The lower winding layer **308** of the transformer is arranged in the same way, and is illustrated in FIG. 4B. The conductive vias are arranged in identical or complementary locations to those in the upper winding layers. However, in the lower winding layer **308** the metallic traces **413**, **423** are formed to connect each outer conductive via **411**, **421** to an inner conductive via **412**, **422** adjacent to the inner conductive via **412**, **422** to which it was connected in the upper winding layer. In this way, the outer conductive vias **411**, **421** and inner conductive vias **412**, **422**, and the metallic traces **413**, **423** on the upper and lower winding layers **308** form coiled conductors around the magnetic core **304**. This is illustrated by way of example in FIG. 4C which shows the connection between adjacent vias in the inner and outer regions by way of the dotted or broken lines. The number of conductive vias allocated to each of the primary and secondary windings determines the winding ratio of the transformer.

[0062] In FIGS. 4A and 4B, optional terminations **440** formed in the substrate **301** of the device are also shown. These may take the form of edge castellations providing for Surface Mount Application (SMA) connections from the device to a printed circuit board on which the device may be mounted.

[0063] In an isolated DC-DC converter, for example, the primary winding **410** and the secondary winding **412** of the transformer must be sufficiently isolated from one another. In FIG. 4A, the central region of the substrate **301**, the region circumscribed by the inner wall of the cavity **302**, forms an isolation region **430** between the primary and the secondary windings. The minimum distance between the inner conductive vias **412** and **422** of the primary and secondary windings **410** and **420** is the insulation distance, and is illustrated in FIG. 4A by arrow **432**.

[0064] Due to the second and the third insulating layers provided in the present preferred embodiment, the distance **432** between the primary and secondary side can be reduced to 0.4 mm, allowing significantly smaller devices to be produced, as well as devices with a higher number of transformer windings. In this context, the spacing between the primary and secondary windings can be measured as the distance between the closest conductive vias **411**, **412** in the

primary side and conductive vias **421**, **422** in the secondary side, and/or between their associated metallic traces.

[0065] The second and third insulating layers need only be on the top and bottom of the device in the central region between the primary and secondary windings. However, in practice it is advantageous to make the second and third insulating layers cover the same area as that of the cover layer **305** and substrate **301** on which they are formed. As will be described below, this provides a support layer for a mounting board on top, and provides additional insulation between the components on that board, and the transformer windings underneath.

[0066] The preferred thickness of the first and second isolation barriers **309a** and **309b** may depend on the safety approval required for the device as well as the expected operating conditions. For example, FR4 has a dielectric strength of around 750 V per mil (0.0254 mm), and if the associated magnitude of the electric field used in an electric field strength test were to be 3000 V, such as that which might be prescribed by the UL60950-1 standard, a minimum thickness of 0.102 mm would be required for the first and second isolation barriers **309a** and **309b**. The thickness of the first and second isolation barriers **309a** and **309b** could be greater than this, subject to the desired dimensions of the final device. Similarly, for test voltages of 1500 V and 2000 V, the minimum thickness of the first and second isolation barriers **309a** and **309b**, if formed of FR4 would be 0.051 mm and 0.068 mm respectively.

[0067] Although solder resist may be added to the exterior surfaces of the second and third insulating layers, i.e., the first and second isolation barriers **309a** and **309b**, this is optional in view of the insulation provided by the layers themselves. FIG. 5 shows optional first solder resist layer **310a** and optional second solder resist layer **310b** with dashed lines. The first solder resist layer **310a** and the second solder resist layer **310b** cover the first isolation barrier **309a** and the second isolation barrier **309b**.

[0068] Although in the preferred embodiment described above, the substrate **301**, cover layer **305**, and the first and second isolation barriers **309a** and **309b** are made of FR4, any suitable PCB laminate system having a sufficient dielectric strength to provide the desired insulation may be included. Non-limiting examples include FR4-08, G11, and FR5.

[0069] As well as the insulating properties of the materials themselves, the cover layer **305** and the insulating layer **309** must bond well with the substrate **301** to form a solid bonded joint. The term solid bonded joint means a solid consistent bonded joint or interface between two materials with little voiding. Such joint should keep its integrity after relevant environmental conditions, for example, high or low temperature, thermal shock, humidity, and so on. Well-known solder resist layers on PCB substrates cannot form such solid bonded joint, and therefore the cover layer **305** and insulating layer **309** are different from such solder resist layers. For this reason, the material for the extra layers is preferably the same as the substrate **301**, as this improves bonding between them. The cover layer **305**, the insulating layer **309**, and the substrate **301** could however be made of different materials providing there is sufficient bonding between them to form a solid body. Any material chosen would also need to have good thermal cycling properties so as not to crack during use and would preferably be hydrophobic so that water would not affect the properties of the device.

[0070] In other preferred embodiments, the insulating substrate **301** could be formed from other insulating materials, such as ceramics, thermoplastics, and epoxies. These may be formed as a solid block with the magnetic core **304** embedded inside. As before, cover layer **305** and first and second isolation barriers **309a** and **309b** would then be laminated onto the substrate **301** to provide the additional insulation.

[0071] The magnetic core **304** is preferably a ferrite core as this provides the device with the desired inductance. Other types of magnetic materials, and even air cores, that is an unfilled cavity formed between the windings of the transformer are also possible in alternative preferred embodiments. Although, in the examples above, the magnetic core is circular in shape, it may have a different shape in other preferred embodiments. Non-limiting examples include, an oval or elongate toroidal shape, a toroidal shape having a gap, EE, EI, I, EFD, EP, UI and UR core shapes. In the present example, a round core shape was found to be the most robust, leading to lower failure rates for the device during production. The magnetic core **304** may be coated with an insulating material to reduce the possibility of breakdown occurring between the conductive magnetic core **304** and the conductive vias **307** or metallic traces **308**. The magnetic core **304** may also have chamfered edges providing a profile or cross section that is rounded.

[0072] Furthermore, although the embedded magnetic component device illustrated above uses conductive vias **307** to connect the upper and lower winding layers **308**, in alternative preferred embodiments, other connections could be used, such as conductive pins. The conductive pins could be inserted into the through holes **306** or could be preformed at appropriate locations in the insulating substrate **301** and cover layer **305**.

[0073] In this description, the terms top, bottom, upper, and lower are used only to define the relative positions of features of the device with respect to each other and in accordance with the orientation shown in the drawings, that is with a notional z axis extending from the bottom of the page to the top of the page. These terms are not therefore intended to indicate the necessary positions of the device features in use, or to limit the position of the features in a general sense.

Second Preferred Embodiment

[0074] A second preferred embodiment will be described with reference to FIG. 5.

[0075] In the first preferred embodiment, the lower winding layer of the transformer primary windings **410** and secondary windings **412** preferably is formed directly on the lower side of the insulating substrate **301**, and the second isolation barrier **309b** is subsequently laminated onto the insulating substrate **301** over the lower winding layer **308**.

[0076] In the second preferred embodiment, the structure of the device **300a** preferably is identical to that described in FIGS. 3A-3F, but in the step illustrated in FIG. 3C, before the through holes **306** are formed, an additional layer, a fourth insulating layer or second cover layer **305b**, is laminated onto the insulating substrate **301**. The through holes **306** are then formed through the substrate **301**, and the first insulating layer **305a** and fourth insulating layer **305b**, and the through holes **306** are plated to form conductive vias **307**. Thus, as illustrated in FIG. 5, in this preferred embodiment, when the lower winding layer **308** is formed, in the

step previously illustrated in FIG. 3E, it is formed on the second cover layer **305b**, rather than the on the lower side of the insulating substrate **301**.

[0077] The second cover layer **305b** provides additional insulation for the lower winding layer **308**.

Third Preferred Embodiment

[0078] In addition to significantly improving the electrical insulation between the primary and secondary side windings of the transformer, the first and second isolation barriers **309a** and **309b** usefully serve as the mounting board on which additional electronic components can be mounted. This allows insulating substrate **301** of the embedded magnetic component device to act as the PCB of more complex devices, such as power supply devices. In this regard, power supply devices may include DC-DC converters, LED driver circuits, AC-DC converters, inverters, power transformers, pulse transformers, and common mode chokes, for example. As the transformer component is embedded in the substrate **301**, more board space on the PCB is available for the other components, and the size of the device can be made small.

[0079] A third preferred embodiment of the present invention will therefore now be described with reference to FIG. 6. FIG. 6 shows example electronic components **501**, **502**, **503** and **504** surface mounted on the first and second isolation barriers **309a** and **309b**. These components may include one or more resistors, capacitors, and switching devices, such as transistors, integrated circuits, and operational amplifiers, for example. Land grid array (LGA) and Ball Grid Array components may also be provided on the first and second isolation barriers **309a** and **309b**.

[0080] Before the electronic components **501**, **502**, **503**, and **504** are mounted on the mounting surface, a plurality of metallic traces are formed on the surfaces of the first and second isolation barriers **309a** and **309b** to make suitable electrical connections with the components. The metallic traces **505**, **506**, **507**, **508**, and **509** are formed in suitable positions for the desired circuit configuration of the device. The electronic components **501**, **502**, **503**, and **504** can then be surface mounted on the device and secured in place by reflow soldering, for example. One or more of the surface mounted components **501**, **502**, **503**, and **504** preferably connects to the primary windings **410** of the transformer, while one or more further components **501**, **502**, **503**, and **504** preferably connects to the secondary windings **420** of the transformer.

[0081] The resulting power supply device **500** shown in FIG. 6 may be constructed based on the embedded magnetic component devices **300** and **300a** shown in FIG. 3F or 5 for example.

Fourth Preferred Embodiment

[0082] A fourth preferred embodiment will now be described with reference to FIG. 7. The embedded magnetic component of FIG. 7 is identical to that of FIGS. 3F and 5 except that further insulating layers are provided on the device. In FIG. 7, for example additional metallic traces **612** are formed on the first and second isolation barriers **309a** and **309b**, and fifth and sixth insulating layers **610a** and **610b** are then formed on the metallic traces **612**. As before, the fifth and sixth insulating layers **610a** and **610b** can be secured to the first and second isolation barriers **309a** and **309b** by lamination or adhesive. Alternatively to being

formed on the first and second isolation barriers **309a** and **309b**, the fifth and sixth insulating layers **610a** and **610b** may be provided by constructing the first and second isolation barriers **309a** and **309b** to have a plurality of layers, such that the fifth and sixth insulating **610a** and **610b** layers are part of the first and second isolation barriers **309a** and **309b**.

[0083] The fifth and sixth insulating layers **610a** and **610b** provide additional depth in which circuit lines can be constructed. For example, the metallic traces **612** can be an additional layer of metallic traces to metallic traces **505**, **506**, **507**, **508**, and **509**, allowing more complicated circuit patterns to be formed. Metallic traces **505**, **506**, **507**, **508**, and **509** on the outer surface can be taken into the inner fifth and sixth insulating layers **610a** and **610b** of the device and back from it, using conductive vias. The metallic traces **505**, **506**, **507**, **508**, and **509** can then cross under metallic traces appearing on the surface without interference. The inner fifth and sixth insulating layers **610a** and **610b** therefore allow extra tracking for the PCB design to aid thermal performance, or more complex PCB designs. The device shown in FIG. 7 may therefore advantageously be used with the surface mounting components **501**, **502**, **503**, and **504** shown in FIG. 6.

[0084] Alternatively, or in addition, the metallic traces of the fifth and sixth insulating layers **610a** and **610b** may be used to provide additional winding layers for the primary and secondary transformer windings. In the examples discussed above, the upper and lower windings **308** are formed on a single level. By forming the upper and lower winding layers **308** on more than one layer it is possible to put the metallic traces of one layer in an overlapping position with another layer. This means that it is more straightforward to take the metallic traces to conductive vias in the interior section of the magnetic core, and potentially more conductive vias can be incorporated into the device.

[0085] Only one of two additional insulating layers **610a** or **610b** may be necessary in practice. Alternatively, more than one additional insulating layer **610a** or **610b** may be provided on the upper or lower side of the device. The additional insulating layers **610a** and **610b** may be used with any of the first, second, or third preferred embodiments.

[0086] In all of the devices described, an optional solder resist cover may be added to the exterior surfaces of the device, either the first and second isolation barriers **309a** and **309b** or the fifth and sixth insulating layers **610a** and **610b**.

[0087] Example preferred embodiments of the present invention have been described for the purposes of illustration only. These are not intended to limit the scope of protection as defined by the attached claims. Features of one preferred embodiment may be used together with features of another preferred embodiment.

[0088] It should be understood that the foregoing description is only illustrative of the present invention. Various alternatives and modifications can be devised by those

skilled in the art without departing from the present invention. Accordingly, the present invention is intended to embrace all such alternatives, modifications, and variances that fall within the scope of the appended claims.

1. A method of manufacturing an embedded magnetic component device, comprising:

- a) preparing an insulating substrate formed of a resin material and including a cavity therein;
- b) installing a magnetic core made of ferrite in the cavity with an air gap between the cavity and the magnetic core, the magnetic core including a first section and a second section;
- c) forming isolated primary and secondary electrical windings that pass through the insulating substrate and that are disposed around the first and second sections of the magnetic core, the secondary winding being spaced away from the primary electrical winding so as to be isolated from the primary electrical winding, the primary electrical winding being located on first and the second sides of the insulating substrate, and the secondary electrical winding being located on the first and the second sides of the insulating substrate;
- d) forming a first isolation barrier of a resin material which is not a solder resist on the first side of the insulating substrate, covering at least the portion of the first side of the insulating substrate between the primary electrical winding and the secondary electrical winding where the primary electrical winding and the secondary electrical winding are closest, to form a solid bonded joint with the first side of the insulating substrate; and
- e) forming a second isolation barrier of a resin material which is not a solder resist on the second side of the insulating substrate, covering at least the portion of the second side of the insulating substrate between the primary electrical winding and the secondary electrical winding where the primary electrical winding and the secondary electrical winding are closest, to form a solid bonded joint with the second side of the insulating substrate; wherein:

step d) includes:

- forming upper conductive traces on the first side of the insulating substrate;
- forming lower conductive traces on the second side of the insulating substrate;
- forming through holes through the insulating substrate;
- plating the through holes to form conductive vias to connect the upper conductive traces and the lower conductive traces and to form a coil conductor around the magnetic core; and
- the first isolation barrier covers the upper conductive traces; and
- the second isolation barrier covers the lower conductive traces.

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